Yuyang Dong

From Data to Models to Agents: Optimizing the Whole System

Initial S | AI Research Leader

Invited Associate Professor, The University of Osaka • Japan

dongyuyang.github.io • GitHub

Profile & Vision

AI researcher and entrepreneur connecting **data systems, foundation models, and AI agents**. At Initial S, building on technical leadership of **Sarashina at SB Intuitions** and contributions to **cotomi at NEC**. Research in data discovery, multimodal understanding, and LLMs provides the foundation for building AI systems that turn information into action.

Vision: Optimize the AI system as a whole—from data quality, memory, and retrieval, through model learning and reasoning, to agent planning and execution. Use feedback from real tasks to improve these layers jointly, balancing task success, reliability, latency, and cost.

Entrepreneurship, Research & Appointments

Initial S Jul 2026 – Present

- Build AI systems that connect personal data, foundation models, and agents.
- Lead company and technology strategy around **end-to-end system optimization**, linking memory and retrieval with model reasoning, agent execution, and feedback.

The University of Osaka May 2026 – Present

Invited Associate Professor

SB Intuitions Sep 2025 – Jun 2026

Chief Research Scientist; LLM Team Tech Lead

- Led the development of **Sarashina**, advancing Japanese-native LLMs through training from scratch.
- Provided technical leadership across LLM pretraining, fine-tuning, and reinforcement learning.

NEC Corporation Apr 2019 – Sep 2025

Generative AI Center / Data Science Research Laboratories

- **Principal Researcher** (Apr 2024 – Sep 2025): R&D on LLMs, VLMs, and multimodal RAG; contributed to **cotomi-v2** development through pretraining data and model evaluation; developed visual document transcription techniques and worked on VLM pretraining and fine-tuning.
- **Special Researcher** (Apr 2021 – Mar 2024): R&D on LLMs and RAG; contributed to **cotomi**, NEC's Japanese LLM, through pretraining data and model evaluation.
- **Researcher** (Apr 2019 – Mar 2021): developed research on table discovery, data lakes, and data enrichment—the data foundation for downstream AI systems.

Education

University of Tsukuba, Japan 2016 – 2019
Ph.D. in Computer Science

University of Tsukuba, Japan 2014 – 2016
Master of Engineering in Computer Science

China University of Mining & Technology, Beijing, China 2013
Bachelor's degree in Computer Science

Selected Publications & Research Impact

Paper titles link to the publication or manuscript. *Equal contribution.

1. **Adaptive Layer Selection for Layer-Wise Token Pruning in LLM Inference**
   Rei Taniguchi, **Yuyang Dong**, Makoto Onizuka, Chuan Xiao.
   Findings of ACL, 2026, pp. 967–986.

2. **SCAN: Semantic Document Layout Analysis for Textual and Visual Retrieval-Augmented Generation**
   Nobuhiro Ueda*, **Yuyang Dong***, Krisztián Boros, Daiki Ito, Takuya Sera, Masafumi Oyamada.
   Findings of EACL, 2026, pp. 1618–1637. **Impact:** Semantic document segmentation improved textual and visual RAG by up to 9.4 and 10.4 points, respectively, in the paper's evaluation.

3. **Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing**
   Haochen Zhang*, **Yuyang Dong***, Chuan Xiao, Masafumi Oyamada.
   EMNLP, 2024, pp. 8754–8782. **Impact:** Co-first-authored open 7B–13B models for local data preprocessing, competitive with GPT-3.5/4 on evaluated tasks.

4. **DeepJoin: Joinable Table Discovery with Pre-trained Language Models**
   Yuyang Dong, Chuan Xiao, Takuma Nozawa, Masafumi Enomoto, Masafumi Oyamada.
   Proceedings of the VLDB Endowment, 16(10), 2023, pp. 2458–2470. **Impact:** First-authored neural table retrieval achieving up to 100× faster search than compared methods with GPU acceleration in the paper's experiments.

5. **Table Enrichment System for Machine Learning**
   Yuyang Dong, Masafumi Oyamada.
   SIGIR, 2022, pp. 3267–3271; system demonstration.

6. **Efficient Joinable Table Discovery in Data Lakes: A High-Dimensional Similarity-Based Approach**
   Yuyang Dong, Kunihiro Takeoka, Chuan Xiao, Masafumi Oyamada.
   ICDE, 2021, pp. 456–467.

7. **Aggregate Reverse Rank Queries**
   Yuyang Dong, Hanxiong Chen, Kazutaka Furuse, Hiroyuki Kitagawa.
   DEXA, 2016, pp. 87–101; **Best Paper Award**.

Selected Preprint

Learning Deliberately, Acting Intuitively: Unlocking Test-Time Reasoning in Multimodal LLMs
Yahan Yu, **Yuyang Dong**, Masafumi Oyamada. arXiv:2507.06999, 2025.

Honors & Awards

DBSJ Kambayashi Young Researcher Award 2022
President's Award, University of Tsukuba 2019
Best Paper Award, DEXA — *Aggregate Reverse Rank Queries* 2016

Research Outreach & Tutorials

Sarashina: Building Japanese-Native LLMs from Scratch
ACM Multimedia Asia, Industry Day, 2025.

Large Language Models for Tabular Data Processing
Database Lecture Series #37, 2025.

On the Use of Large Language Models for Table Tasks
CIKM tutorial, 2024. With Masafumi Oyamada, Chuan Xiao, and Haochen Zhang.

Academic Service & Languages

Organization: Poster & Encore Track Co-Chair, ADMA 2025.

Reviewing: TKDE, KAIS, IEICE, IEEE Access, IPSJ; conference reviewing includes ICML (2019–2020), NeurIPS (2019–2020), and AAAI (2020–2021).

Languages: Chinese (native); Japanese and English (professional working proficiency).